EXPERIMENT 1

Academic Year: 2026-27 Year and Sem: TE V
Subject: DEV Lab Course Code: 2175113

PROBLEM STATEMENT:

To study the Agile Software Development Lifecycle and create a kanban project using Jira.

SOFTWARE REQUIREMENTS:

Internet Connection ,Google Chrome / Firefox,Jira Cloud Account ,Email ID

THEORY:

What is Agile?
Agile is a software development methodology where software is developed in small, manageable parts called **iterations or sprints**. Instead of completing the entire project at once, Agile delivers working software continuously.

Agile Principles

- Customer collaboration
- Continuous improvement
- Working software
- Frequent delivery
- Team communication
- Adaptability to change

Agile Lifecycle

Requirement Gathering
↓
Planning
↓
Design
↓
Development
↓
Testing
↓
Deployment
↓

Feedback
↓
Next Sprint

What is Kanban?

Kanban is an Agile framework that visualizes work on a board and focuses on continuous delivery rather than fixed sprints.

- Visualize workflow
- Limit Work in Progress (WIP)
- Manage workflow continuously
- Continuous improvement

What is Jira?

Jira is an Agile project management tool developed by Atlassian.
It helps teams to

- Manage projects
- Track bugs
- Create user stories
- Plan sprints
- Assign tasks
- Monitor project progress

Practical Example

Project: Library Management System
Epic: Book Management

Story	Tasks
Story 1: Add New Book As a librarian, I want to add a new book so that it is available in the catalogue.	• Design Add Book Form • Validate Book Details • Save Book Information
Story 2: Search Book As a student, I want to search for a book so that I can quickly find its availability.	• Design Search Page • Implement Search Function • Display Search Results

Procedure

Step 1: Create a Jira Cloud account.
Step 2: Create a new Kanban project.
Step 3: Project Name: Library Management System.
Step 4: Create one Epic named 'Book Management'.

Step 5: Create two Stories: Add New Book and Search Book.
Step 6: Create tasks under each story.
Step 7: Assign tasks to team members.
Step 8: Move issues through Backlog → To Do → In Progress → Review → Done.
Step 9: Observe the Kanban board and monitor progress.
Step 10: View Control Chart and Cumulative Flow Diagram reports.

Kanban Workflow

Backlog → To Do → In Progress → Review → Done

CONCLUSION : Successfully implemented the Agile Software Development Lifecycle using Jira by creating and managing a Kanban project. The experiment provided hands-on experience in organizing work using a Kanban Board, creating and managing issues such as Epics, User Stories, Tasks, and Bugs, prioritizing the backlog, assigning tasks to team members, and tracking work through different workflow stages (To Do, In Progress, Done). Students also learned to monitor project progress using Jira dashboards and reports, gaining practical exposure to continuous workflow management, task visualization, and collaborative software development using the Kanban methodology.